Lab 3-2 Network Address Translation

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- **Translation of addresses between networks (NAT).**
- **Configuration of Easy IP.**

Topology

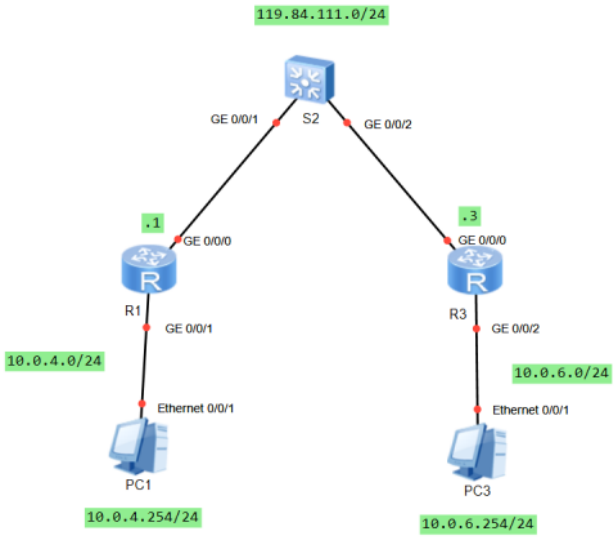


Figure 3.2 Network Address Translation Topology

Scenario

In order to conserve addressing the offices of the enterprise network have implemented private addressing internally. Users however require a means to be routed between these private networks and the public network domain. R1 and R3 represent edge routers of the enterprise branch offices ,the branch network need access to the public network. The administrator of the network is requested to configure dynamic NAT solutions on the in order to allow R1 to perform address translation. An

easyIP NAT solution is to be applied to R3.

Tasks

Step 1 Preparing the environment.

```
[Huawei]sysname R1
[Rl]inter GigabitEthernet0/0/1
[Rl-GigabitEthernet0/0/1]ip address 10.0.4.1 24

[Huawei]sysname R3
[R3]interface GigabitEthernet0/0/2
[R3-GigabitEthernet0/0/2]ip address 10.0.6.3 24
```

Set ip addresses, subnet masks and default gateways on PC1 and PC3.

Step 3 IP configuration

```
[Rl]interface GigabitEthernet0/0/0
[Rl-GigabitEthernet0/0/0]ip address 119.84.111.1 24

[R3]interface GigabitEthernet0/0/0
[R3-GigabitEthernet0/0/0]ip address 119.84.111.3 24

[S2]interfae vlanif1
[S2-Vlanif1]ip address 119.84.111.2 24
```

Verify that R1 is able to reach both PC1 and R3.

```
<Rl>ping 10.0.4.254
  PING 10.0.4.254: 56 data bytes, press CTRL_C to break
    Reply   from10.0.4.254: bytes=56 Sequence=l ttl=255 time=23 ms
    Reply   from10.0.4.254: bytes=56 Sequence=2 ttl=254 time=l ms
    Reply   from10.0.4.254: bytes=56 Sequence=3 ttl=254 time=l ms
    Reply   from10.0.4.254: bytes=56 Sequence=4 ttl=254 time=10 ms
    Reply   from10.0.4.254: bytes=56 Sequence=5 ttl=254 time=l ms
   -  10.0.4.254 ping statistics --
     5 packet(s) transmitted
     5 packet(s) received
     0.00% packet loss

     round-trip min/avg/max = 1/7/23 ms
```

```
<Rl>ping 119.84.111.3
PING 119.84.111.3: 56 data bytes, press CTRL_C to break
Reply from 119.84.111.3: bytes=56 Sequence=l ttl=255 time=l ms
Reply from 119.84.111.3: bytes=56 Sequence=2 ttl=255 time=10 ms
Reply from 119.84.111.3: bytes=56 Sequence=3 ttl=255 time=l ms
Reply from 119.84.111.3: bytes=56 Sequence=4 ttl=255 time=l ms
Reply from 119.84.111.3: bytes=56 Sequence=5 ttl=255 time=10 ms

 ---  119.84.111.3 ping statistics --
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 1/4/10 ms
```

Step 4 Configure Access Control Lists for R1 and R3.

Configure an advanced ACL on R1 and select the data flow with the source of SI, the destination of R3, and destined for the telnet service port.

```
[Rl]acl 3000
[Rl-acl-adv-3000]rule 5 permit tcp source 10.0.4.254 0.0.0.0 destination
119.84.111.3  0.0.0.0 destination-port eq 23
[Rl-acl-adv-3000]rule 10 permit ip source 10.0.4.0 0.0.0.255 destination any
[Rl-acl-adv-3000]rule 15 deny ip
```

Configure a basic ACL on R3 and select the data flow whose source IP address is 10.0.6.0/24.

```
[R3]acl 2000
[R3-acl-basic-2000]rule permit source 10.0.6.0 0.0.0.255
```

Step 5 Configure Dynamic NAT.

Configure dynamic NAT on the GigabitEthernetO/O/O interface of R1.

```
[R1]nat address-group 1 119.84.111.240 119.84.111.243
[Rl]interface GigabitEthernet 0/0/0
[Rl-GigabitEthernetO/O/O]nat outbound 3000 address-group 1
```

Verify the address group has been configured correctly

```
<Rl>display nat address-group
    NAT Address-Group Information:

    Index Start-address       End-address

    1      119.84.111.240   119.84.111.243
```

```
     Total : 1
```

Test connectivity to the gateway of the remote peer from the internal network.

```
<PCl>ping 119.84.111.3
  PING 119.84.111.3: 56 data bytes, press CTRL_C to break
    Request time out
    Reply from  119.84.111.3:bytes=56  Sequence=2 ttl=254 time=l ms
    Reply from  119.84.111.3:bytes=56  Sequence=3 ttl=254 time=l ms
    Reply from  119.84.111.3:bytes=56  Sequence=4 ttl=254 time=l ms
    Reply from  119.84.111.3:bytes=56  Sequence=5 ttl=254 time=l ms

   -- 119.84.111.3 ping statistics  -
    5 packet(s) transmitted
    4 packet(s) received
    20.00% packet loss
    round-trip min/avg/max = 1/1/1 ms
```

Do not exit the telnet session, instead open a second session window to R1 and view the results of the ACL and NAT session translation.

```
<Rl>display acl 3000
Advanced ACL 3000, 2 rules
Acl's step is 5
rule 5 permit tcp source 10.0.4.254 0 destination 119.84. Ill. 3 0 destination-port
eq telnet (1 matches)
rule 10 permit ip source 10.0.4.0 0.0.0.255 (1 matches)
rule 15 deny ip

<Rl>display nat session all
  NAT Session Table Information:

    Protocol                 ICMP(1)
    SrcAddr Vpn              10.0.  4.254
    DestAddr Vpn             119.84.111.3
    Type Code Icmpld         8   0   44003
    NAT-Info
      New SrcAddr            119.84.111.242
      New DestAddr
      New Icmpld             10247

    Protocol                 TCP(6)
    SrcAddr Port Vpn         10.0.
                             4.254          49646
    DestAddr Port Vpn        119.84.111.3    23
    NAT-Info
      New SrcAddr            119.84.111.242
      New SrcPort            10249
      New DestAddr
      New DestPort
```

```
  Total : 2
```

The ICMP session has a lifetime of only 20 seconds and therefore may not appear to be present when displaying the NAT session results. The following command can be used in this case to extend the period over which the ICMP results are maintained:

```
[Rl]firewall-nat session icmp aging-time 300
```

Configure easyIP on the Gigabit Ethernet 0/0/0 interface of R3, associating the easyIP configuration with ACL 2000 that had been configured earlier.

```
[R3-GigabitEthernet0/0/0]nat outbound 2000
```

Test the connectivity from PC3 to R1 via R3.

```
<PC3>ping 119.84.111.1
PING 119.84.111.1: 56 data bytes, press CTRL C to break
Reply from 119.84.111.1: bytes=56  Sequence=l ttl=254 time=l ms
Reply from 119.84.111.1: bytes=56  Sequence=2 ttl=254 time=l ms
Reply from 119.84.111.1: bytes=56  Sequence=3 ttl=254 time=l ms
Reply from 119.84.111.1: bytes=56  Sequence=4 ttl=254 time=l ms
Reply from 119.84.111.1: bytes=56  Sequence=5 ttl=254 time=l ms

   -- 119.84.111.1 ping statistics  --
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 1/1/1 ms

<R3>display acl 2000
Basic ACL 2000, 1 rule
Acl's step is 5
rule 5 permit source 10.0.6.0 0.0.0.255 (1 matches)

<R3>display nat outbound acl 2000

NAT Outbound Information:

Interface                 Acl          Address-group/IP/Interface
                                      Type

GigabitEthernet0/0/0 2000                      119.84.111.3 easyip
```